

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
A																	
B																	
C																	
D																	
E																	
F																	
G																	
H																	
I																	
J																	
K																	
L																	

LOGO SIGNAGE SYSTEM

Bangkok 10330

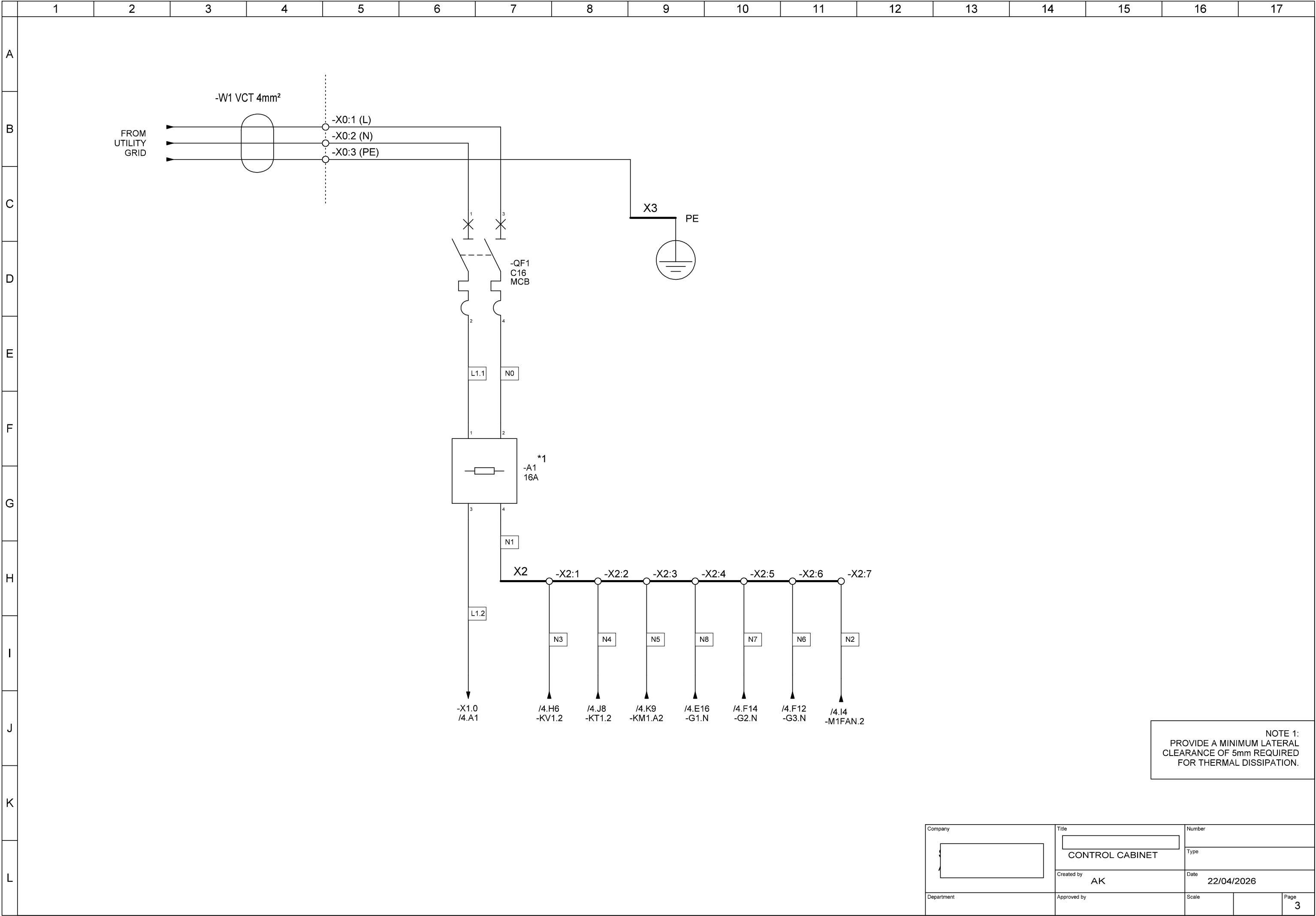
7 PAGES

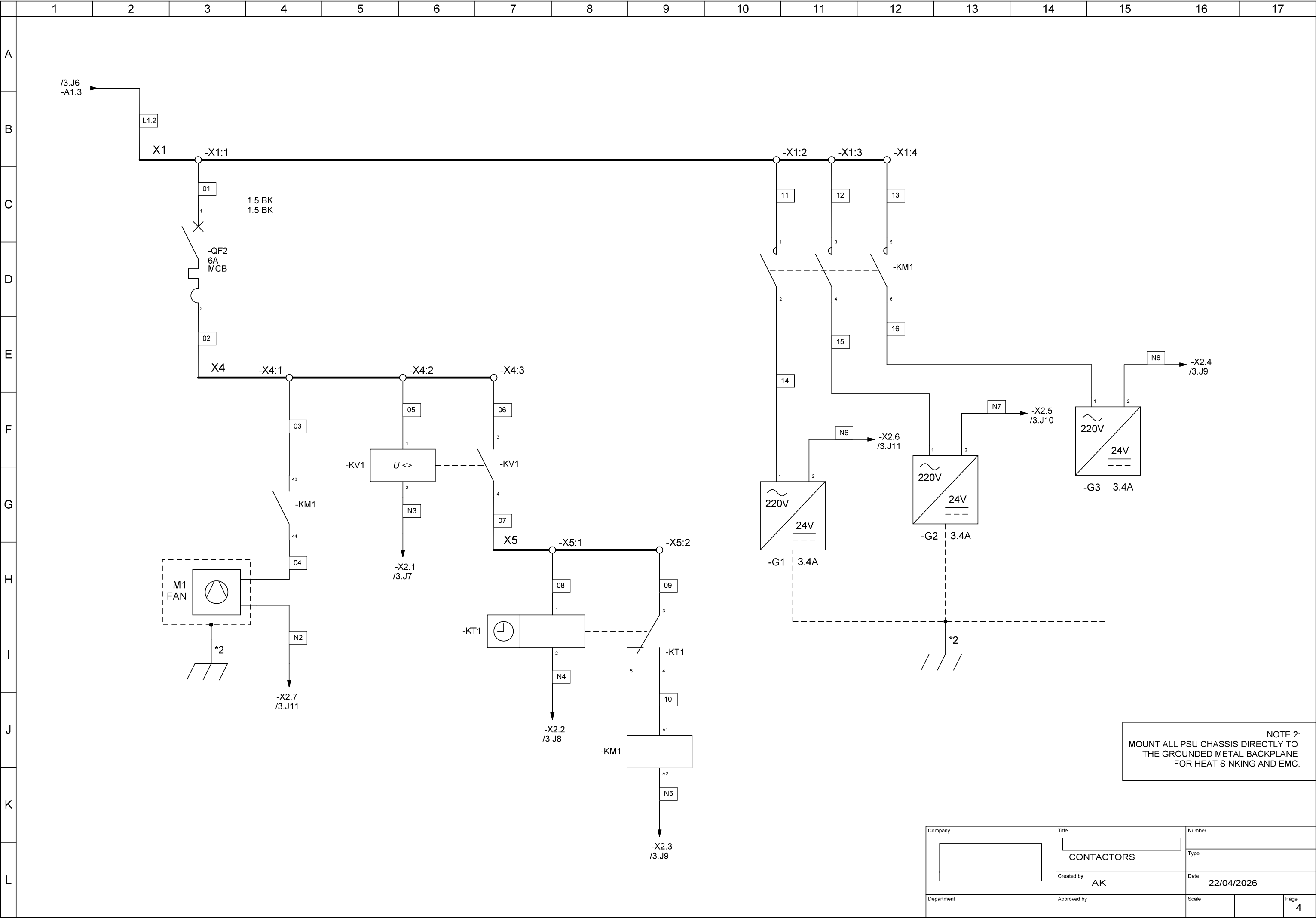
Company 	Title TITLE	Number		
	Created by AK	Date 22/04/2026		
	Approved by	Scale		Page 1

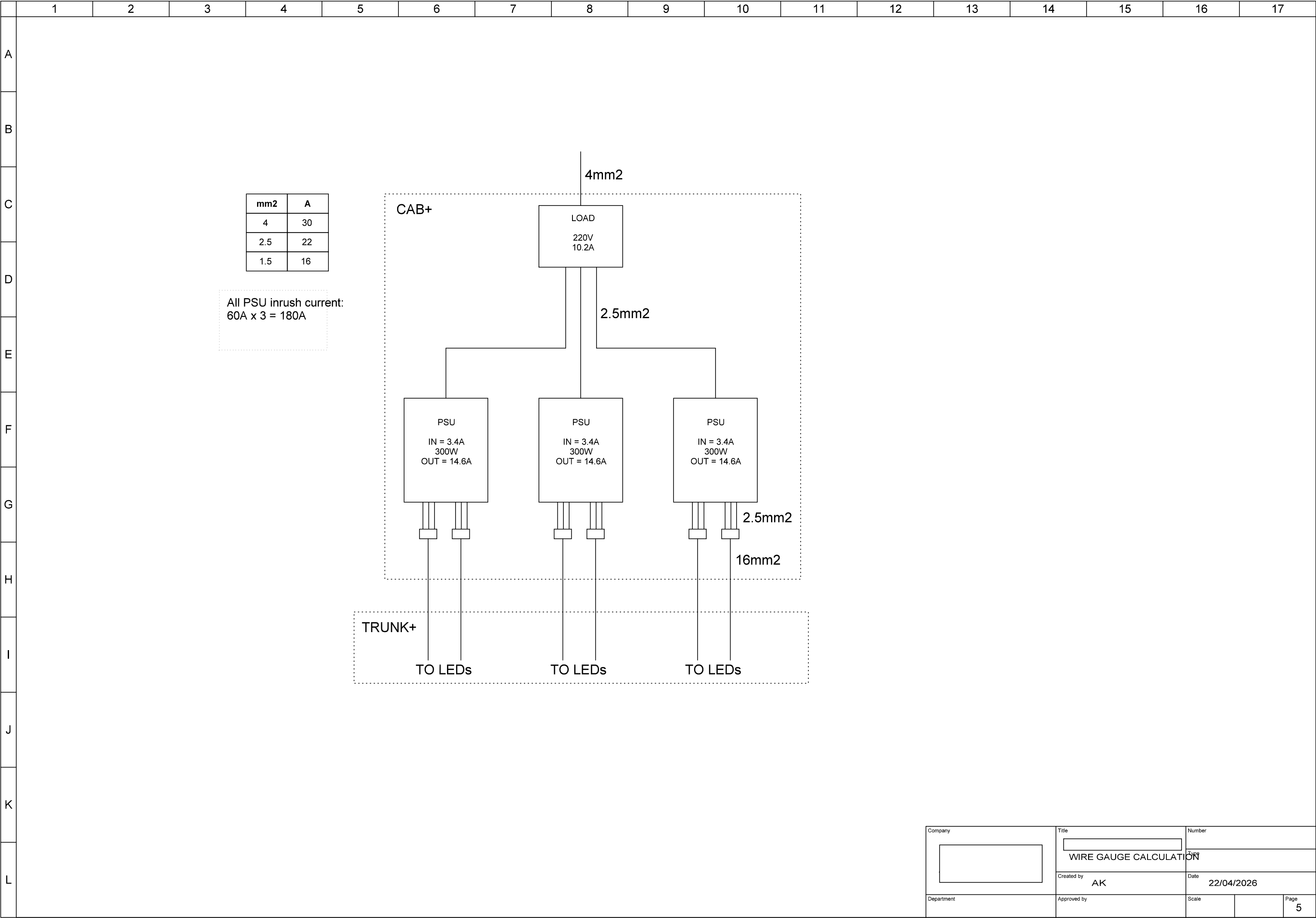
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
A																	
B																	
C																	
D																	
E																	
F																	
G																	
H																	
I																	
J																	
K																	
L																	

IDENTIFIER	COMPONENT NAME	VALUE	PART NAME	COMMENT
QF1	CIRCUIT BREAKER TYPE-C 16A MCB	C16	ABB Circuit Breaker S202M-C16 2P-16A	
QF2	CIRCUIT BREAKER TYPE-C 6A MCB	C6	ABB Circuit Breaker SH201-C6 1P-6A	
A1	INRUSH CURRENT LIMITER	16A	MeanWell AC Inrush Current Limiter ICL-16R	
X0	TERMINAL (3P)	41A	BUK-5N (0.2-4mm2) 41A 800V	
X1	TERMINAL (6P + 1S)	32A	BUK-2.5B (0.2-2.5mm2) 32A 800V	
X2	TERMINAL (9P + 2S)	32A	BUK-2.5B (0.2-2.5mm2) 32A 800V	
X3	TERMINAL PE-TYPE	32A	BUK-2.5B (0.2-2.5mm2) 32A 800V	
X4	TERMINAL (4P + 1S)	32A	BUK-2.5B (0.2-2.5mm2) 32A 800V	
X5	TERMINAL (3P + 1S)	32A	BUK-2.5B (0.2-2.5mm2) 32A 800V	
X1, X4, X5	TERMINAL BUS (3 PCS)	32A	BUK-2.5B (0.2-2.5mm2) 32A 800V	
X6, X7, X8	TERMINAL BLOCK 4x4	40A	UKK80 Unipolar Distribution Terminal Block	x6
M1 FAN	ENCLOSE COOLER FAN	220V	SAKAGUCHI Enclosure Fan S8025HA2BL 220VAC	
KV1	VOLTAGE MONITORING RELAY	220V	Voltage Monitoring Relay RS PRO 122-215	
KT1	TIMER CLOCK	220V	Digital Timer THC15A 220VAC	
KM1	CONTACTOR 22A	AC COIL	Contactor LG GMC-22 AC-Coil 22A	
G, G2, G3	POWER SUPPLY	24V	MeanWell LRS-350-24 350W 24V	x3
---	WIRE DUCT	---	WIRE DUCT 30x30mm NANO	3m
---	CABLE GLAND	---	PG-25 BLACK	x7
---	CABLE GLAND	---	PG-16 BLACK	x2
---	CRIMP TERMINAL	---	VE1508	1 pack
---	CRIMP TERMINAL	---	VE2508	1 pack
---	CRIMP TERMINAL	---	VE16-12	1 pack
---	CRIMP TERMINAL FORK	---	SV2-4	1 pack
---	WIRE 1.5mm2 RED	---	VSF-YZ-1.5	15m
---	WIRE 2.5mm2 BROWN	---	VSF-YZ-2.5	15m
---	WIRE 2.5mm2 BLUE	---	VSF-YZ-2.5	15m
---	WIRE 2.5mm2 PE	---	VSF-YZ-2.5	15m
---	WIRE 16mm2	---	SS VCT-2x16	122m

Company	Title	Number		
		Type		
	BILL OF MATERIALS			
	Created by AK	Date 22/04/2026		
Department	Approved by	Scale		Page 2







A

B

C

D

E

F

G

H

I

J

K

L

LINEAR POWER DISSPATION CALCULATION (IEC)

$$V_{drop} = I \times \left(0.0175 \times \frac{2L}{A}\right)$$

$$P_{heat} = I \times V_{drop}$$

$$S = \pi \times d \times L$$

$$\Delta T = \frac{P}{\alpha \times S}$$

Where:
I = Current (Ampers). Constant (14.6A)
L = Length of Wire (Meters)
A = Wire Cross Section (Millimeters)

Where:
P(Heat) = Power (Watts);
L = Current (Ampers)
V(drop) = Voltage Drop (Volts)

Where:
S = Total Wire Surface Area (Meters)
π = Constant (3.14159)
L = Length of Wire (Meters)
d = External Diametr of Wire (Meters)
*if this is 1-core cable - multiply S by x2

Where:
ΔT = Temperature Rise (Degree, 'C)
P = Power (Watts)
L = Length of Wire (Meters)
S = Total Wire Surface Area (Meters)
α = Heat Transfer Coefficient (10)

Cable (Wire) Type	Cross Section, mm2	Wire Length, m	Voltage Drop, V	Power Heat, W	Watt per 1 meter of wire	d, m	S, m	ΔT, 'C
LAPP H07V-K 1x10	10	34	1.7	25	0.73	0.0062	1.324	1.8
LAPP H07V-K 1x16	16	34	1.1	16.06	0.5	0.0081	1.73	0.93
LAPP H07V-K 1x25	25	34	0.7	10.1	0.3	0.0098	1.05	0.96

A

B

C

D

E

F

G

H

I

J

K

L

LINEAR POWER DISSPATION CALCULATION (IEC)

$$V_{drop} = I \times \left(0.0175 \times \frac{2L}{A}\right)$$

Where:
I = Current (Ampers). Constant (14.6A)
L = Length of Wire (Meters)
A = Wire Cross Section (Millimeters)

$$P_{heat} = I \times V_{drop}$$

Where:
P(Heat) = Power (Watts);
L = Current (Ampers)
V(drop) = Voltage Drop (Volts)

$$S = \pi \times d \times L$$

Where:
S = Total Wire Surface Area (Meters)
 π = Constant (3.14159)
L = Length of Wire (Meters)
d = External Diametr of Wire (Meters)
*if this is 1-core cable - multiply S by x2

$$\Delta T = \frac{P}{\alpha \times S}$$

Where:
 ΔT = Temperature Rise (Degree, 'C)
P = Power (Watts)
L = Length of Wire (Meters)
S = Total Wire Surface Area (Meters)
 α = Heat Transfer Coefficient (10)

Cable (Wire) Type	Cross Section, mm2	Wire Length, m	Voltage Drop, V	Power Heat, W	Watt per 1 meter of wire	d, m	S, m	ΔT , 'C
LAPP H07V-K 1x10	10	50	2.6	38	0.76	0.0062	1.948	2.0
LAPP H07V-K 1x16	16	50	1.6	23.36	0.4	0.0081	2.5	1.52
LAPP H07V-K 1x25	25	50	1.02	15	0.3	0.0098	3.08	0.5